
Response Geometry in Integrated Gradients: A Diffusion-Policy Case Study

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Abstract

Attribution rankings select inputs for interventions, and their evaluations also need responses that score those interventions. We study this distinction for Integrated Gradients (IG) attributions of an action chunk predicted by a diffusion policy with fixed noise. For quadratic and stabilized negative- ℓ_2 action-discrepancy targets, we give an exact relation between their normalized responses. We also derive a positive, path-dependent factor between their gradients. Using a smooth construction, we show that IG rankings can reverse while the nonzero gradients remain pointwise collinear. A finite-grid counterexample shows that 0.5 cannot serve as a universal random-ranking AUC reference. Reanalyzing 99 preserved Robotics Diffusion Transformer files, we find that changing only the evaluation response changes median vision deletion AUC from 0.4479 to 0.2908 although the recorded quadratic-target rankings stay the same. In all eight paired rescoring cases, every median response change is positive although six means are negative because of overshooting tails. Exploratory episode-omission results retain insertion-mean signs but can reverse language-deletion signs. Another smooth construction gives several integration grids with the same completeness, ranking and perturbation curves but an ordering that reverses the exact IG order. In this example, the inaccurate ordering scores better on both perturbation metrics. Diagnostics on the actual model separate repeatability from unresolved integration-budget sensitivity in an incomplete cohort. Together, the results separate response-geometry effects from attribution-ranking quality and numerical-integration accuracy. They do not qualify a production numerical setting, demonstrate better attribution rankings, or explain task success or dependence on learned weights. The artifact retains raw-record lineage, analytic checks, adverse results and provenance limits.

1 Introduction

Vision-language-action (VLA) models like RT-2 (Zitkovich et al., 2023), Octo (Octo Model Team et al., 2024), and the Robotics Diffusion Transformer (RDT) (Liu et al., 2025) use robot observations along with language instructions to predict robot actions.¹ Diffusion policies (Chi et al., 2023) iteratively remove noise (Ho et al., 2020) to produce action chunks. Applying a scalar attribution method to this vector output requires deciding which aspect of the action prediction to explain. This decision also affects how the explanation is evaluated.

Integrated Gradients (IG) integrates the gradient of a scalar output along a path between a baseline and an input (Sundararajan et al., 2017). For a fixed-noise action sample, the action chunk’s discrepancy from a detached reference can define this scalar function. A quadratic discrepancy and a negative norm both measure change, but they are different scalar functions. Evaluating the attribution also requires a response that scores interventions. The ranking selects which features to remove or insert, while the response measures the effect of those changes. If both the scalar target and evaluation response change, native-response AUC alone cannot determine whether the ranking has improved.

We analyze these issues through exact derivations, constructive examples and a retrospective RDT case study. For fixed selected features and perturbed outputs, changing the response from a quadratic discrepancy to

¹AI tools were used to develop and check mathematical constructions, design and implement saved-data analyses, review code and literature, and draft and revise this manuscript. The author is responsible for the content.

a stabilized negative ℓ_2 discrepancy changes normalized AUC. At the same time, the positive scaling factor between the targets’ gradients varies along the path, which allows their IG rankings to differ. Rescoring interventions is therefore a distinct analysis from recomputing attributions. Neither result shows which ranking is more useful when the response is held fixed.

We also ask whether the computed attribution accurately approximates the specified mathematical quantity. Small scalar completeness residuals can hide errors in individual input coordinates. The same inaccurate IG approximation can also appear when a finite-precision calculation is repeated. The smooth construction in Appendix B gives nested grids that agree exactly on completeness, ranking and perturbation responses while exact IG has the reversed ranking. Our diagnostics report the checks already collected together with the full planned roster and its unfinished cells. Passing checks in part of this roster does not certify the implementation. The intended attributions measure action discrepancy at a fixed context. They do not explain eventual task success or ensure safe robot behavior.

We make four contributions.

1. **Response and path-weight analysis.** We derive an identity for normalized responses with precise qualifications for overshoot and zero-gap cases. We also give a constructive example with a finite stabilizer in which the quadratic and negative-norm targets reverse IG feature order.
2. **A finite-grid calibration counterexample.** A policy with 100 equally contributing features has quadratic-response insertion and deletion AUC equal to 0.65976 for all orderings on the historical evaluation grid. This shows why 0.5 is not a universal chance reference.
3. **An authenticated retrospective case study.** We rescore saved RDT perturbation curves while holding feature rankings fixed. The analysis records the occurrence rules, episode grouping and links to the underlying source records. Unknown historical context identities prevent a paired comparison of ranking quality.
4. **Numerical diagnostic limits.** We give a smooth counterexample to certification by finite-grid agreement and report an incomplete campaign of fixed-input repeatability and integration-budget checks. Check failures and incomplete coverage remain visible. We do not certify settings for production use.

Together, these contributions separate the issues and document their consequences in one diffusion-policy family. A ranking study that holds the response fixed would address the additional question of ranking efficacy. The absence of that study limits what we can conclude about efficacy, but it does not invalidate the analysis that holds rankings fixed.

2 Related Work

2.1 Attribution Methods

Sundararajan et al. (2017) introduced IG with the sensitivity and implementation-invariance axioms, demonstrating that straight-line path integration satisfies both. Subsequent work addresses limitations of the original method. Jha et al. (2021) computed attributions over neural stochastic differential equation models for smoother maps. Their later work improved robustness by shaping the injected noise (Jha et al., 2022). Walker et al. (2024) showed that gradients in saturation regions add noise and proposed Integrated Decision Gradients, which puts more of the integration on the part of the path relevant to the model’s decision. For vision transformers, metric-driven attributions (Walker et al., 2025b) and information-flow decompositions (Walker et al., 2025a) improve patch-level explanations. Guided IG (Kapishnikov et al., 2021) changes the integration path to reduce noise in pixel-level attributions.

Several works analyze sensitivity to the integration budget. Makino et al. (2024) examine instance-varying completeness residuals, nonmonotone changes with step count and unresolved gradients along the IG path. Singhi & Kumari (2024) analyze step-count sensitivity in IG variants and the Important Direction Gradient Integration (IDGI) framework. Their reported stability experiment compares JPEG-compressed and

uncompressed inputs. Our diagnostics instead keep the context fixed and analyze repeatability separately from changes in attribution coordinates across integration budgets. Appendix B gives a smooth construction that shows a limit of these checks. Agreement on finitely many integration grids together with exact scalar completeness does not imply accurate coordinate integrals.

Petsiuk et al. (2018) proposed the insertion and deletion curves that we use as ranking-based faithfulness metrics. Several works analyze the reliability of perturbation metrics. Hooker et al. (2019) evaluate attributions by ablating inputs and retraining the model. Rong et al. (2022) identify information leakage through the shape of the mask when deletion is performed without retraining. Tomsett et al. (2020) find common saliency metrics to be unreliable. This literature already establishes that the evaluation depends on the metric. Our case study examines this dependence for action-discrepancy responses through an algebraic relationship between normalized curves and by rescaling preserved interventions with their feature ordering fixed. Adebayo et al. (2018) introduced model and data randomization tests to check whether saliency maps depend on the model’s trained parameters and training data. These checks show that even visually plausible attributions do not establish that dependence. Our backbone randomizations are limited because the learned encoders and adaptors are kept in place. For image diffusion models, cross-attention maps relate generated pixels to specific prompt tokens (Tang et al., 2023). This is different from attributing the scalar discrepancy of a predicted control chunk.

Our implementation uses IG with modality-specific baselines to attribute a fixed-noise action-discrepancy function. Path weighting is already used in methods such as IDG (Walker et al., 2024). The factor we derive follows specifically from changing the scalar discrepancy target. We apply these established concerns to quadratic and stabilized negative-norm action-discrepancy functions and examine their separate effects on fixed interventions and integrated rankings.

2.2 Vision-Language-Action Models

VLA models combine perception, language understanding and action generation. RT-2 (Zitkovich et al., 2023) fine-tunes a vision-language model to predict actions that are discretized and represented as token sequences. Octo (Octo Model Team et al., 2024) is a transformer-based generalist policy trained on a large multi-robot dataset. Diffusion Policy (Chi et al., 2023) formulates robot action generation as conditional denoising diffusion (Ho et al., 2020). Action chunks are generated through iterative refinement instead of single-step regression. RDT (Liu et al., 2025) scales this diffusion-based action prediction using a Diffusion Transformer (DiT) backbone, a SigLIP image encoder (Zhai et al., 2023), and a T5-XXL text encoder (Raffel et al., 2020). Each frame from three cameras is encoded through SigLIP. A history of two frames gives six camera slots and produces 4,374 vision tokens. These are combined with up to 1,024 language tokens and a 128-dimensional unified state vector. The model then generates chunks of 64 action steps with a five-step denoising schedule using DPM-Solver++ (Lu et al., 2025). The multiple modalities and repeated denoising steps make RDT an interesting but challenging model for attribution.

2.3 Interpretability in Robotics

Earlier work on robot-policy explanations focused on reinforcement learning agents in discrete-action settings. Greydanus et al. (2018) studied perturbation-based saliency in the Atari domain. Counterfactual methods (Amitai et al., 2024) explain policies by comparing what happens under executed actions with what would happen under alternatives. VisualBackProp (Bojarski et al., 2018) visualizes convolutional neural network (CNN) saliency for end-to-end driving policies in continuous control. These are convolutional driving models rather than multimodal diffusion samplers. Recent VLA methods examine different quantities to explain. Zhang et al. (2026) estimate visual influence through randomized perturbations and action-discrepancy responses, with explicit assumptions relating mean-squared error to distributional divergence. VLA-Trace (Shi et al., 2026) combines representation-level comparisons, attention interventions and behavioral probes. IMPACT-VLA (Kim & Park, 2026) attributes phase-modality interventions to downstream task success through counterfactual closed-loop execution. Our estimand is the discrepancy of a predicted action chunk under fixed noise and at a fixed context. It does not explain eventual task success.

We focus on how response geometry affects the evaluation of IG rankings for this estimand and make no claim to originate VLA interpretability.

3 Attribution Targets and Evaluation

3.1 How Integrated Gradients Is Computed

For a differentiable function $F: \mathbb{R}^n \rightarrow \mathbb{R}$, input $x \in \mathbb{R}^n$, and baseline $x' \in \mathbb{R}^n$, Integrated Gradients follows the straight-line path from the baseline to the input. The contribution of dimension i is computed by integrating the gradient with respect to that dimension,

$$\text{IG}_i(x, x') = (x_i - x'_i) \int_0^1 \frac{\partial F}{\partial x_i} \Big|_{x' + \alpha(x - x')} d\alpha. \quad (1)$$

Under conditions where the fundamental theorem of calculus and the chain rule apply along the path, IG obeys *completeness*. This means that $\sum_i \text{IG}_i(x, x') = F(x) - F(x')$. We approximate the integral as a uniform Riemann sum with m steps,

$$\text{IG}_i \approx \frac{x_i - x'_i}{m+1} \sum_{r=0}^m \frac{\partial F}{\partial x_i} \Big|_{x' + \frac{r}{m}(x - x')}. \quad (2)$$

In the historical implementation, all terms receive equal weight, including both endpoints of the integration path. Convergence of this approximation to exact IG depends on how well-behaved the function being integrated is along the path. Summation with finite numerical precision creates an additional source of error. Increasing m does not distinguish this error from integration error or necessarily make it smaller.

3.2 Keeping the Target Fixed Through Denoising

At policy call t , the diffusion sampler output is $g_\theta(o_t, z_t)$, a function of the observation o_t and fixed diffusion noise z_t . We write this predicted action chunk as μ_t to maintain consistency with historical records. It is a sample obtained with fixed diffusion noise. This notation alone does not identify it as a conditional mean. We compute the reference $a_t = g_\theta(o_t, z_t)$ once without gradients. Under an intervention, we use the auxiliary attribution target

$$Q_t = -\frac{1}{2\sigma^2 D} \|a_t - \mu_t\|^2, \quad (3)$$

where $\sigma^2 = 1$ and $D = 512$ active entries cover 64 predicted time steps over eight active dimensions. The score is the portion of a Gaussian log kernel that depends on the action difference, averaged over entries and excluding the log-normalizing constant. It has not been derived as the conditional log likelihood of the diffusion policy. Historical code and files refer to this target as `logpi`, and its change is written as ΔQ . While the execution loop acts only on a subsampled prefix of the chunk before stopping or calling the policy again, we compute attributions for the whole predicted chunk.

We keep the noise realization fixed across points along the integration path so that the target remains a deterministic function of the input. An expectation over noise would be a different quantity to attribute. Explaining it would require justification for switching the order of the gradient and expectation, along with a Monte Carlo treatment. Using a new independent noise sample at every path point does not compute the fixed-noise quantity studied here. In the historical implementation, nonfinite gradient components were replaced by zeros. This could make individual IG coordinates wrong even when their errors cancel in the sum. That cancellation can leave the scalar completeness residual small or zero. Passing a completeness check therefore does not mean that each coordinate is correct.

3.3 Other Scalar Targets

We compare the quadratic target with two other targets for the action difference, given by

$$y_t^{\ell_2} = -\sqrt{\|a_t - \mu_t\|_2^2 + \epsilon}, \quad y_t^{\max} = -\max_j |a_{t,j} - \mu_{t,j}|, \quad (4)$$

For our analyses, the stabilizer used in the implementation is $\epsilon = 10^{-12}$. Both targets use the same D active entries. The stabilized ℓ_2 target is a smooth function of the residual and has zero gradient where the predicted and reference actions are identical. Away from the stabilizer scale, its gradient with respect to the actions approaches unit norm. The maximum-deviation target, which we call maxdev, measures the action difference using a different norm. It is not generally a monotone transformation of ℓ_2 . For example, the residual vectors $(1, 1)$ and $(1.1, 0)$ reverse order under these norms. For a fixed perturbation, the quadratic and stabilized ℓ_2 responses have an algebraic relationship, but their gradients can still give different attribution rankings. Differences in scale or units do not imply that a target with a larger magnitude contains more information or is more useful. Section 5.1 focuses on response changes between the quadratic and stabilized ℓ_2 targets. Records for maxdev remain available in the released analysis as an additional diagnostic.

3.4 How the Response Changes the Weight Along the Path

We fix the reference action, active coordinates, model, noise and other modalities. We then define $r(x) = \|a - \mu(x)\|_2^2$ and $L(x) = -\sqrt{r(x) + \epsilon}$. By the chain rule, wherever the gradients exist,

$$\nabla Q(x) = \frac{\sqrt{r(x) + \epsilon}}{D\sigma^2} \nabla L(x). \quad (5)$$

The nonzero gradients of the quadratic and stabilized targets therefore have the same direction at each point. Both targets' gradients are zero when the prediction exactly matches its own reference. These facts alone do not determine what happens when the gradients are integrated. Along a straight-line path $x(\alpha) = x' + \alpha(x - x')$, the integrated gradient is

$$\text{IG}_i^Q = (x_i - x'_i) \int_0^1 \frac{\sqrt{r(x(\alpha)) + \epsilon}}{D\sigma^2} \partial_i L(x(\alpha)) d\alpha. \quad (6)$$

The weights are positive but generally vary with α . Different coordinates can contribute more in different regions of the path. The change in weighting can therefore change the ranking after integration. This does not conflict with IG implementation invariance, which compares different implementations of the same scalar function (Sundararajan et al., 2017).

For an example, consider the policy with scalar output $\mu(x_1, x_2) = x_1^2 + \frac{4}{5}x_2$, baseline $(0, 0)$, actual input $(1, 1)$, detached reference $a = \frac{9}{5}$, and $D = \sigma^2 = 1$. Performing the integration exactly along the same straight-line path gives

$$\text{IG}^Q = \left(\frac{23}{30}, \frac{64}{75} \right), \quad \lim_{\epsilon \downarrow 0} \text{IG}^L = \left(1, \frac{4}{5} \right). \quad (7)$$

The quadratic target ranks feature 2 first. The stabilized L target with $\epsilon = 10^{-12}$ ranks feature 1 first, by a margin of at least 0.1999975. Appendix A proves this bound using the finite stabilizer. The reversal therefore holds for the implemented response and does not rely on replacing it with an idealized limiting value. This shows that rankings can reverse in a smooth nonlinear policy. It does not show that this happens for all such policies or that it occurs empirically in RDT.

3.5 Attribution at Each Policy Call

RDT predicts a chunk of 64 action steps at a time, subsampled $4\times$ to 16 steps in the environment. IG is performed once per *policy call*, since all actions in the chunk result from a single decision. We attribute one policy query rather than each environment step. The wall-clock runtime of IG depends on implementation details, hardware, numerical precision and the integration budget. We do not use the earlier timing summaries as a performance result.

At each policy call, we perform attribution separately on each set of modality inputs, using a corresponding forward function that fixes the remaining inputs.

- **Vision.** Adapted vision tokens are interpolated from baseline to real representations while language and state inputs are fixed.

Algorithm 1 IG at each policy call with fixed noise. In Section 4, we describe which historical artifacts are available and their limits.

Input. Policy π_θ , environment $\mathcal{E}$, integration steps m , target $y \in \{Q, \ell_2, \text{maxdev}\}$

Output. Attributions $\{A_t^V, A_t^L, A_t^S\}$ at every policy call.

```

1:  $o_0 \leftarrow \mathcal{E}.\text{reset}()$ 
2: for policy call  $t = 0, 1, \dots$  do
3:   Encode  $o_t$  into vision  $v_t$ , language  $l_t$ , state  $s_t$ 
4:   Draw and store  $z_t$ . Compute  $a_t \leftarrow g_\theta(o_t, z_t)$  without gradients
5:   Define  $F$  from  $a_t$ , the fixed-noise predicted chunk  $\mu_t$ , and the chosen target
6:   for modality  $M \in \{V, L, S\}$  do
7:     Set baseline  $x'_M$ , input  $x_M$ 
8:     Freeze the other two modalities
9:      $A_t^M \leftarrow \text{IG}(F_M, x_M, x'_M, m)$ 
10:  end for
11:  Record  $A_t^V, A_t^L, A_t^S$  and completeness errors
12:  Apply action chunk and advance  $\mathcal{E}$  by up to 16 steps, stopping on termination
13: end for

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Eq. 2

- **Language.** Adapted language tokens are interpolated from a minimal two-token baseline encoded through T5-XXL to the real instruction embedding, while vision and state inputs are fixed.
- **State.** The 128-dimensional unified state vector is interpolated from zeros to its real value while vision and language inputs are fixed. For the Franka arm, only eight dimensions are active, covering the seven joints and the gripper. We include the state-adaptor multilayer perceptron (MLP) inside the forward function to preserve the joint structure.

Algorithm 1 summarizes how these attributions are computed.

Choosing the baseline. The vision baseline is a gray image in the external-camera slot, with a uniform value. The remaining five camera slots contain constant background images and stay fixed at their real values. IG therefore attributes only to the slot that contains information about the scene. The language baseline is a minimal two-token representation using the tokenizer’s padding and end-of-sequence tokens. There is no beginning-of-sequence token in T5. Rather than taking a zero vector as the baseline, we encode these tokens through T5-XXL. The encoder output is then padded with zeros to the stored embedding length. The language forward passes attend to the real token span for the task. This gives an attended baseline consisting of the encoded token pair, with zeros in the remaining positions being attended to. The GELU-based language adaptor allows a zero vector as an input, but that would be a different representation baseline. The preliminary effectiveness comparisons cannot be independently reconstructed. For state, the baseline is a zero vector in normalized joint space. This choice does not establish that it represents a physically supported state or is useful as an explanation reference. Appendix C.2 reports the available checks of sensitivity to the vision baseline.

3.6 How We Evaluate the Attributions

Completeness. For each modality, we hold the other inputs fixed and compute the absolute residual $|\sum_i \text{IG}_i - (F(x) - F(x'))|$. We also compute its relative version by dividing by the magnitude of the scalar endpoint gap,

$$\text{Err} = \frac{|\sum_i \text{IG}_i - (F(x) - F(x'))|}{|F(x) - F(x')|}. \quad (8)$$

This relative value is undefined when the signed gap is exactly zero. If the gap is small but nonzero, division can amplify numerical error. The historical action-norm filter $\|a_t\| \geq 15$ does not bound this denominator. We retain the filter only to show how changing the included population affects the results. A small residual indicates agreement in the scalar accounting identity. It does not rule out errors in individual coordinates that cancel in the sum but still change the attribution map or feature ranking.

Perturbation curves. We sum the absolute coordinate attributions within each eligible feature group and rank the groups by these sums. For vision, the groups are patch tokens from the external camera. For language, they are the attended instruction tokens. Deletion sets the features of the first groups in the ranking to their baseline values. Insertion starts from the baseline input and inserts the original features for the first groups in that ranking. For a scalar response R , the normalized value is

$$\tilde{R}(x_k) = \frac{R(x_k) - R(x')}{R(x) - R(x')}. \quad (9)$$

We integrate on the fraction grid stored in each record. The normalized curves and their AUCs are not necessarily within $[0, 1]$. The historical grid requests $\{0, 1, 5, 10, 20, 30, 50, 75, 100\}\%$ of the eligible feature groups. For language, rounding the token counts changes the fractions actually used. The saved curves should therefore not be interpreted as continuous functions sampled at evenly spaced points. Top- k score changes are $R(x_k) - R(x)$, so they have the units of the scalar response R . We distinguish the attribution target used for ranking from the response used to score the interventions. The revised tables do not use the earlier study-specific thresholds as tests of effectiveness.

Randomization and baseline checks. For C1, we reinitialize model parameters, compute attributions again and correlate them with the originals. The standard version changes the final action-output projection and recomputes the reference action. The frozen-reference and backbone-cascade versions instead retain the original reference. The historical cascade randomly reinitializes all backbone parameter matrices of rank at least two simultaneously. It retains one-dimensional parameters, encoders and adaptors. It does not randomize the layers progressively in a cascade. For C2, we shuffle external-camera pixels as well as language. If a cache of shuffled pre-T5 language is available, the historical implementation uses it. Otherwise, it permutes the post-T5 embeddings. The saved records do not establish which mode was used for a given attribution call. These tests measure different forms of dependence. C2 is not the label-randomization test proposed by Adebayo et al. (2018). The historical implementation computes perturbed maps at $m = 16$, while the original maps nominally use $m = 64$. C1 therefore affects the numerical approximation as well as the model weights. To check baseline sensitivity, we compare black, gray and blurred observation features.

4 Retrospective Data and Statistical Protocol

Scope and traceability. The revised analysis checks 99 preserved files containing 22,966 valid records and 1 specifically identified corrupt line. The released inventory identifies each valid physical record by source path, original line number and byte hash. Some recorded keys occur more than once. There are 129 such groups, and 58 differ in scientific fields as well as, or instead of, timing metadata. The last occurrence is used as a declared rule for this retrospective analysis. When records conflict, this rule does not establish that the selected attempt is correct. A separate sensitivity analysis excludes every key with a conflicting payload to show how those conflicts affect the results. The released lineage registry maps records and occurrence decisions to population membership, result identifiers and generated manuscript cells. These links support checks of the preserved bytes. They cannot recover missing checkpoints, observations, action tensors or noise.

Campaigns and models. The main reconstructed RDT-170M tables come from the saved Month 7 verification campaign at $m = 64$, covering PickCube, StackCube, PegInsertionSide and PickSingleYCB in ManiSkill3 (Tao et al., 2024). These experiments are separate from the original main pass, whose inputs are unavailable. The verification campaign used pretrained RDT-170M, while the saved RDT-1B PickCube campaign used a task-finetuned checkpoint. The comparison therefore mixes changes in training, trajectories, historical preprocessing and parameter count. It cannot isolate the effect of model scale or competence. Appendix C consequently reports the 1B cohort separately. The primary tables exclude missing original completeness and standard-sanity results, along with missing early PickCube/StackCube integration-budget results.

Known protocol limits. Historical attribution covers the full predicted action chunk, although the simulator executes only a subsampled chunk. The four-task selection also reflects historical implementation

decisions. An attempted TurnFaucet run used the wrong controller and has no documented corrected rerun. Its unsuccessful outcomes therefore do not establish that the model lacked the capability for that task. Early follow-up jobs replaced encoded padding/end tokens with a zero language baseline, and the PickSingleYCB follow-up used a proxy PickCube instruction embedding. The pooled 1B language cohort consequently mixes the early baseline with later encoded baselines. Earlier sidecar filenames omitted seed and integration-budget fields, allowing files to be overwritten and possibly reused across nominal runs. The seed-tagged Month 7 records reduce this filename problem but cannot recover the old contexts. Matching scalar fingerprints alone therefore cannot establish exact context equality.

Estimands and uncertainty. The primary population includes all valid saved calls after the declared occurrence rule is applied. Continuous call-level quantities are summarized by medians over calls. Residual pass rates are proportions over the stated call population. Solver contrasts give relative changes between endpoint medians, while the separately named equal-episode analysis uses episode medians. Uncertainty summaries resample whole recorded episode groups, formed from task, model, recorded seed and episode, with $B = 10,000$ draws and seed zero. Each point estimate and its interval use exactly the same source-record population. The percentile intervals describe the released empirical mixture of episodes. Longer episodes contribute more calls to that mixture. These intervals do not measure variability across independently trained checkpoints. The separately named 1B sensitivity instead gives each episode equal weight through the median of episode medians. For the old randomization variants, we also join nominal task/episode keys across recorded seeds when reuse is possible. This checks conditional sensitivity to dependence between those records. Neither grouping recovers unobserved context identity. Very small cluster counts are stated explicitly. Resampling two baseline groups, in particular, is not evidence of calibrated interval coverage. All unfiltered results and the historical norm-selected sensitivity remain available. Tails are not removed to obtain a favorable threshold verdict.

The paired rescoring analysis keeps each call’s ranking and interventions fixed. It defines $\Delta_{ej} = \text{AUC}_{Q,ej} - \text{AUC}_{L,ej}$ for call j in recorded episode e . The primary estimand is $E^{-1} \sum_{e=1}^E n_e^{-1} \sum_{j=1}^{n_e} \Delta_{ej}$, giving every episode equal weight. A prespecified descriptive sensitivity uses the median of paired call differences. Both summarize within-call changes. Neither is the difference between the marginal medians reported above. The protocol specified all eight ranking/modality/direction combinations and the same 10,000-draw episode-bootstrap convention before these summaries were computed. Its intervals are conditional descriptions without a multiplicity adjustment. They do not establish calibrated joint coverage or global significance. If either response has a zero endpoint gap, the paired difference remains undefined and stays in the accounting. All 750 calls in each ranking cohort have defined paired differences, and each of the 30 recorded episodes contributes 25 calls. The equal-episode and equal-call means therefore coincide here.

5 Response Geometry and Retrospective Results

5.1 Separating Attribution Target from Evaluation Response

A recorded intervention can be scored under another response without recomputing IG or changing any selected feature. Write the recorded active-action discrepancy as d , with $Q = -d^2/1024$, and $L = -\sqrt{d^2 + \epsilon}$ with $\epsilon = 10^{-12}$. The two responses are related by

$$L = -\sqrt{-1024Q + \epsilon}, \quad Q = -\frac{L^2 - \epsilon}{1024}. \quad (10)$$

Under exact self-reference at the actual endpoint, if the all-baseline squared residual is $R > 0$, the normalized values satisfy

$$\begin{aligned} N_Q(r) &= 1 - \frac{r}{R}, & N_L(r) &= \frac{\sqrt{R + \epsilon} - \sqrt{r + \epsilon}}{\sqrt{R + \epsilon} - \sqrt{\epsilon}}, \\ N_Q(r) - N_L(r) &= \frac{(\sqrt{r + \epsilon} - \sqrt{\epsilon})(\sqrt{R + \epsilon} - \sqrt{r + \epsilon})}{R}. \end{aligned} \quad (11)$$

Thus $N_Q > N_L$ for $0 < r < R$, with equality at both endpoints and the order reversed for $r > R$. For any curve whose interventions all stay in range, nonnegative integration weights preserve this weak ordering.

Monotonicity is unnecessary. The area inequality is strict only when an interior residual receives positive integration weight. Overshoots can reverse the area ordering and must be retained rather than clipped. When $R = 0$, neither normalization is defined. Changing from Q to L therefore lowers both insertion and deletion AUC for an in-range curve. Larger insertion AUC and smaller deletion AUC are often taken to indicate better ranking quality. Under those conventions, the same response change suggests opposite changes in quality even though the interventions are identical. The historical calculation does not require the recorded actual endpoint to be an exact self-reference. Instead, every saved curve ordinate is transformed, its endpoints are recomputed under the new response, and it is integrated again on the same saved grid. Duplicate handling leaves 750 calls in 30 episodes for each ranking cohort. For quadratic-target vision rankings, the median deletion AUC drops from 0.4479 under Q to 0.2908 under L . For stabilized- ℓ_2 rankings, it rises from 0.2830 under L to 0.4404 under Q . Table 1 and Figure 1 report both modalities and both perturbation directions.

Table 1: The response changes while the saved rankings stay fixed. The table reports median AUC and conditional 95% episode-bootstrap intervals. Each ranking cohort contains 750 calls from 30 episodes and reuses the same recorded interventions. The two ranking cohorts have not been established as paired contexts, so comparing them cannot show that either ranking is better.

Ranking target	Response	Modality	Insertion AUC	Deletion AUC
Q	Q	Vision	0.883 [0.845, 0.909]	0.448 [0.408, 0.504]
Q	Q	Language	0.580 [0.510, 0.658]	0.372 [0.311, 0.428]
Q	L	Vision	0.756 [0.718, 0.785]	0.291 [0.253, 0.333]
Q	L	Language	0.469 [0.412, 0.516]	0.257 [0.228, 0.284]
L	Q	Vision	0.884 [0.833, 0.921]	0.440 [0.402, 0.501]
L	Q	Language	0.635 [0.582, 0.720]	0.368 [0.302, 0.407]
L	L	Vision	0.756 [0.700, 0.803]	0.283 [0.252, 0.327]
L	L	Language	0.507 [0.456, 0.566]	0.249 [0.202, 0.278]

The scores within each saved ranking cohort therefore depend on response geometry. This does not show that Q-IG and L2-IG select the same features, give equivalent rankings or provide effective rankings. The attempted join found different reference-action norms at every nominal key. Exact raw contexts and noise are unavailable as well. Agreement after common rescoring cannot therefore replace a prospective paired comparison of rankings. A larger native L2 score also does not by itself show that information was recovered. Stabilized L2 has no square-root kink at exact agreement.

Paired changes and overshooting tails. Table 2 shows why the summary must be specified. Every one of the eight paired-call medians is positive, from 0.1041 to 0.1590, but six equal-episode means are negative. For Q-ranking vision insertion, the paired-call median is 0.1042, while the equal-episode mean is -17.0966 with conditional interval $[-33.8210, -3.9746]$. The most negative paired change in this case is -4438.8386 . These finite extremes remain in the analysis without clipping or trimming. Appendix D gives an exploratory omission check for all eight cases. All four negative insertion means remain negative after any one recorded episode is omitted. The two negative language-deletion means can change sign under these omissions, consistent with their conditional intervals already crossing zero. This shows how the results are concentrated within the recorded cohort. It does not establish population robustness or replace the full untrimmed primary estimands.

To examine the tails, each saved squared-discrepancy value is classified relative to the interval between its stored actual and baseline endpoint discrepancies. This does not assume exact self-reference. A high-distance overshoot exceeds both endpoints, while a low-distance undershoot falls below both. The artifact records which endpoint is larger and retains both departures, undefined gaps and a prespecified sensitivity to boundary labels. In Q-ranking vision insertion, 312 of 750 calls have at least one high-distance overshoot, and 89 have a negative paired AUC change. Decomposing the trapezoidal difference by saved-node category gives mean contributions of 0.1144 from in-range nodes and -17.2111 from high-distance overshoots. There is no low-distance contribution in this case. The contributions sum to the reported mean. The negative

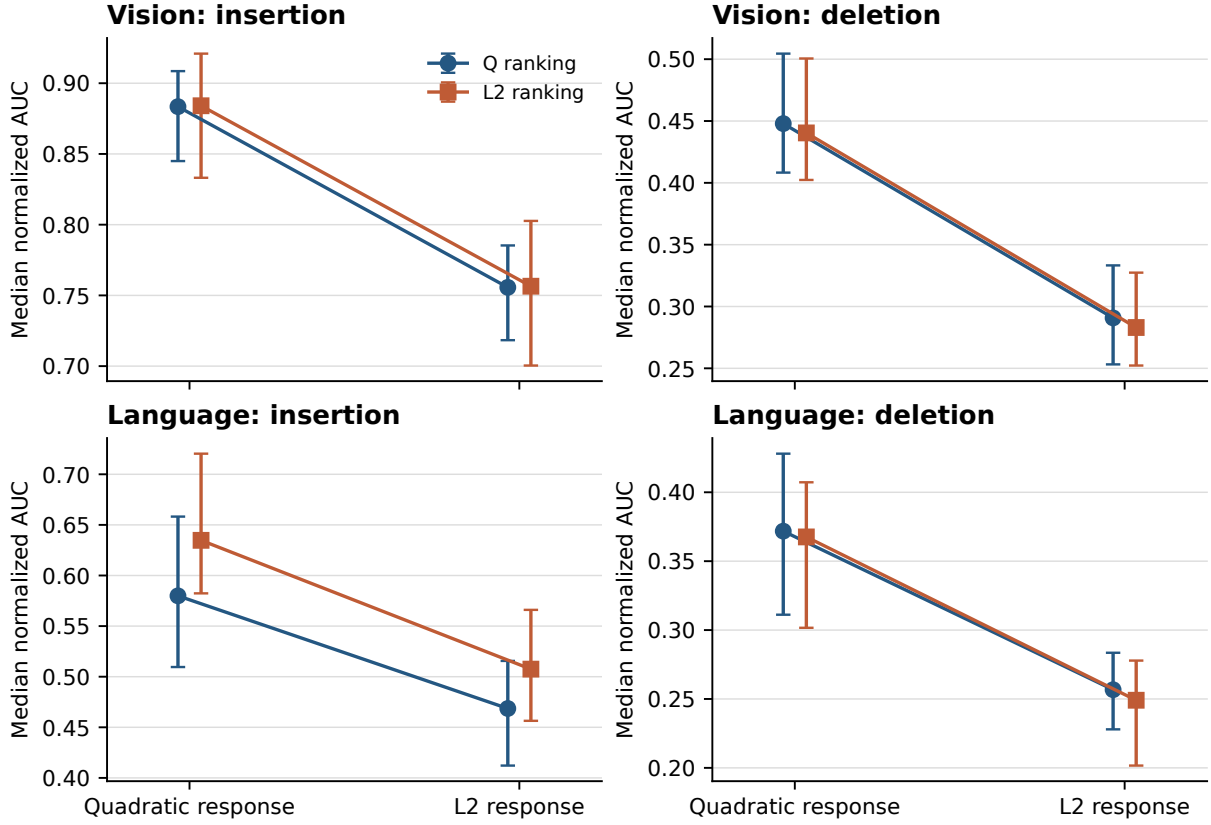


Figure 1: Changing the evaluation response while keeping each recorded ranking and its interventions fixed. Error bars show conditional episode-bootstrap intervals for each median. Lines connect responses within one ranking cohort. The two ranking cohorts are not authenticated pairs, and no universal chance reference line is supplied.

Table 2: Paired response differences $\Delta = \text{AUC}_Q - \text{AUC}_L$ within the same saved curve. Every row uses 750 calls from 30 recorded episodes, and rows within a ranking cohort reuse those calls. The primary mean gives episodes equal weight. Its 95% interval is a conditional, unadjusted episode-bootstrap description. The paired-call median is also included because overshooting tails can reverse the mean’s sign. Neither summary compares the quality of Q and L2 rankings.

Ranking	Modality	Curve	Mean Δ	95% interval	Median Δ
Q	vision	insertion	-17.0966	[-33.8210, -3.9746]	0.1042
Q	vision	deletion	0.1227	[0.1059, 0.1382]	0.1588
Q	lang	insertion	-0.1819	[-0.4051, -0.0277]	0.1169
Q	lang	deletion	-0.0095	[-0.1022, 0.0631]	0.1189
L2	vision	insertion	-18.5620	[-37.8672, -3.1731]	0.1041
L2	vision	deletion	0.0184	[-0.2037, 0.1378]	0.1590
L2	lang	insertion	-0.0329	[-0.0963, 0.0266]	0.1279
L2	lang	deletion	-1.3710	[-4.2081, 0.0780]	0.1205

contribution from recorded overshoots is therefore large enough to reverse its sign. The L2-ranking vision-insertion cohort shows the same qualitative disagreement between mean and median within its own records. Node categories and segment endpoint labels describe sampled interventions only. They do not establish how the continuous curve behaves between those points. The in-range ordering in Equation 11 therefore does not mean that changing the response moves every curve in an empirical population in the same direction.

5.2 An Analytic Random-Order Counterexample

As an analytic control, consider the finite additive scalar policy $f(x) = \sum_{i=1}^n x_i$ at an all-one input, a zero baseline and the self-reference $f(x) = n$. Every feature ordering gives the same perturbation values. Whole-feature prefix interventions realize only the fractions $q = k/n$, with $k = 0, \dots, n$. The normalized quadratic deletion and insertion responses at those fractions are $1 - q^2$ and $1 - (1 - q)^2$. Trapezoidal integration over all $n + 1$ prefix fractions gives

$$\text{AUC}_n = \frac{2}{3} - \frac{1}{6n^2}$$

for either curve. Extending these values as polynomials and integrating continuously over $q \in [0, 1]$ gives $2/3$. This is also the $n \rightarrow \infty$ limit, but it is not the exact finite-grid area. With $n = 100$, every fraction on the historical nine-point requested grid is realized exactly. Trapezoidal integration gives 0.65976 for both curves under every ordering. This finite construction rules out a universal random-order AUC of 0.5. For comparison, the unstabilized negative-norm response is linear on the same fractions and has trapezoidal AUC 0.5. A fixed reference such as 0.5 therefore needs justification for the particular policy, response and intervention protocol. This is an analytic construction, not an empirical measurement of random-ranking performance on RDT.

5.3 Completeness and Perturbation Summaries

Table 3 reports completeness residuals from the saved verification campaign. Within these cohorts, vision and language have smaller median relative residuals than state. This summary does not establish that their coordinates are more accurate or that dimension count causes the difference. The historical integrator combines errors from the model forward pass, path arithmetic, summation and quadrature. Actual-model validation must separate these factors on identical inputs before production settings are chosen.

Table 3: The Month 7 verification campaign uses all valid calls at $m = 64$. Each cell reports the median relative completeness residual as a percentage and a conditional 95% episode-bootstrap interval. The estimate and interval use exactly the same calls and episodes. These historical numerical checks do not establish that individual attribution coordinates are accurate.

Task	Calls	Episodes	Vision (%)	Language (%)	State (%)
PegInsertion	700	100	2.17 [1.94, 2.35]	1.83 [1.69, 1.98]	3.48 [3.00, 4.04]
PickCube	397	100	2.20 [1.90, 2.43]	1.88 [1.62, 2.30]	5.88 [4.99, 7.29]
PickYCB	397	100	1.85 [1.71, 2.06]	2.02 [1.79, 2.18]	4.26 [3.50, 5.62]
StackCube	400	100	2.39 [2.18, 2.59]	1.76 [1.58, 1.97]	10.53 [8.07, 13.13]

Table 4 reports native quadratic-response insertion and deletion AUCs from the same verification campaign. The scores describe recorded perturbations. Without matched random and simple ranking controls, their magnitudes do not establish attribution efficacy. Normalization can produce extreme negative AUCs. The minimum saved PickCube vision insertion AUC in the all-valid population is -18190.36. We keep this observation and its endpoint denominator. Neither clipping the value nor applying the action-norm filter would establish its validity. The released diagnostics include denominator and tail information alongside the medians because a central summary cannot describe the whole curve population.

Table 4: Results from the Month 7 verification campaign, using all valid calls and the native quadratic response. Medians and conditional 95% intervals use the same population. Entries are not classified as successful against an uncalibrated chance threshold. These descriptive estimates retain the limits of the stored token-fraction grids, endpoint conventions and recorded provenance.

Task	Modality	Calls	Episodes	Insertion AUC	Deletion AUC
PegInsertion	Vision	700	100	0.731 [0.688, 0.774]	0.280 [0.234, 0.309]
PegInsertion	Language	700	100	0.320 [0.257, 0.379]	0.341 [0.291, 0.382]
PickCube	Vision	397	100	0.822 [0.771, 0.838]	0.167 [0.137, 0.213]
PickCube	Language	397	100	0.525 [0.468, 0.580]	0.433 [0.391, 0.475]
PickYCB	Vision	397	100	0.844 [0.808, 0.863]	0.337 [0.296, 0.372]
PickYCB	Language	397	100	0.335 [0.227, 0.409]	0.471 [0.438, 0.500]
StackCube	Vision	400	100	0.821 [0.777, 0.849]	0.208 [0.161, 0.261]
StackCube	Language	400	100	0.359 [0.292, 0.432]	0.480 [0.429, 0.512]

6 Numerical Diagnostics from an Incomplete Campaign

The later campaign checks identified current computations separately from the historical records above. Historical runs were rescored arithmetically using saved curves, but their original contexts could not be recovered. The later probe instead evaluates a separately identified offline fp32 function on authenticated contexts with its own reference actions. This function cannot validate the historical attribution maps or provide a paired comparison with them. The campaign was limited by compute cost and stopped before its planned numerical coverage and independent verification were complete. The analysis uses a fixed, authenticated snapshot of sealed results. Entries that were incomplete at the snapshot are neither successful nor failed calculations. The stopping decision supplied no statistical sampling rule for the completed subset. Passed and violated criteria therefore describe only that subset, not population failure rates.

Functions and precision controls. Stored observations, masks, reference actions and exact noise arrays provide fixed, authenticated inputs for comparison. The diagnostic implementation separates accumulation arithmetic, master-path and final-product arithmetic, quadrature, and downstream model precision. It compares full coordinate arrays, ranks over eligible groups, selected groups and perturbation responses. The earlier downstream-fp32 factor control lifts the same bf16-rounded parameters and cached adapted representations. It also converts non-parameter dtype attributes explicitly while holding the original reference action fixed. This changes downstream arithmetic while preserving the numeric inputs. It does not recover unrounded checkpoint parameters or test fp32 encoders. Reusing the original reference also distinguishes this control from the later probe. These earlier comparisons preceded the full repeated-gradient checks, so they cannot by themselves establish a globally repeatable precision contrast.

The later engineering campaign defines its own deterministic fp32 function. Image and language representations retain cached bf16 values lifted to fp32, raw state is re-adapted in fp32, and the reference action is the function’s own fp32 prediction. The function uses contexts visited by the bf16 policy and the stored source noise. It is an offline intervention function, not the bf16 behavior policy or a derivative of that policy’s executed commands. Repeating endpoint outputs checks a different property from repeating path gradients and attribution maps. Differences caused by changing the function or its reference therefore cannot be attributed to quadrature alone.

Engineering design and interpretation. The predeclared engineering bank includes four RDT-170M tasks and two tasks on an independently identified authors’ 1B checkpoint. Identifying this checkpoint does not recover or authenticate the historical task-finetuned 1B checkpoint used in Appendix C. Source collection allowed at most 13 policy calls, with a further cap of 400 environment steps and possible earlier termination. Call indices 0 and 12 were selected in advance. The analysis therefore uses fixed engineering call indices rather than a uniform sample of calls from complete trajectories. Episodes were not selected by success or action norm. The integration ladders use trapezoidal quadrature with fp32 path and accumulation arithmetic. The

same gradient stream produces a mean path-gradient map without IG’s input-difference multiplier. This is a numerical diagnostic, not an evaluated efficacy baseline. The protocol compares coordinates, ranks, completeness and common perturbation responses across integration budgets and repeated calculations. Its tolerances diagnose changes in the tested approximation. They are neither efficacy margins nor calibrated error bounds relative to an exact attribution. The finest-budget map is a comparator, not ground truth. The cohort analysis retains every planned context, including source-rollout calls that were never reached and entries with no sealed result.

Preserved coverage. The retained snapshot is labeled October 1, 2026, 03:36 UTC, before the later compute stop. Table 5 covers all 24 planned contexts and 144 target/modality cells. Twelve contexts have all six diagnostics complete. These represent six episodes across the four 170M tasks and contribute 72 complete cells. Three planned calls were never reached by their source rollouts, accounting for 18 cells. The remaining nine available contexts, or 54 cells, have no sealed diagnostics at this cutoff. No 1B numerical context is complete in the snapshot. These are snapshot statuses. They do not claim the final outcome of work that was later interrupted.

Table 5: The full planned engineering roster at the authenticated 03:36 UTC snapshot. The first four numerical columns count contexts, and the last gives complete/planned target-modality cells. Each context has six cells. Source-rollout calls that were not reached remain separate from available contexts with unsealed results. Complete does not imply numerical qualification.

Stratum	Planned	Complete	Unreached	Unsealed	Cells
PegInsertionSide 170M	4	2	0	2	12/24
PickCube 170M	4	4	0	0	24/24
PickCube 1B	4	0	2	2	0/24
PickSingleYCB 170M	4	2	0	2	12/24
StackCube 170M	4	4	0	0	24/24
StackCube 1B	4	0	1	3	0/24
Total	24	12	3	9	72/144

Repeatability and budget sensitivity. The image and language ladders use $m = 128, 256, 512$, and the state ladder uses $m = 256, 512, 1024$. The finest maps were repeated twice, and seven fixed path locations were probed three times each. Every recorded repeated-condition equality and coverage check met its conditions. These checks use the same finite contexts. They are not independent replications and do not estimate a population reliability rate.

Table 6 reports every completed target/modality group at its two finest recorded budgets. For maps A_m and A_{2m} , the coordinate discrepancy is $\|A_m - A_{2m}\|_1 / \|A_{2m}\|_1$. This quantity is undefined when its denominator is zero. The predeclared relative-coordinate and completeness thresholds are both 1%. Other checks cover group ordering and sensitivity of the common responses. The table counts a violation when a context/target/modality arm exceeds at least one pairwise threshold in this finer-budget comparison, including mean path-gradient diagnostics. Finest-budget completeness is summarized separately. All lower-budget comparisons, raw denominators and individual checks are retained in the artifact. The maxima describe only these recorded cases. They have no intervals and do not support population generalization.

At both recorded PickSingleYCB episode-1 contexts, both targets violate the state completeness threshold at the finest budget, and their two finest maps differ substantially. The scalar gaps exceed the declared near-zero cutoffs. The artifact also retains their signed gaps and absolute residuals. Division by an almost-zero scalar gap therefore cannot explain these discrepancies. In a state case, intervention response curves remain unchanged despite a substantial coordinate change. Response stability alone therefore cannot qualify the integral. The quadratic target also fails, so endpoint behavior of the stabilized negative-norm target cannot explain all failures. Sparse stored path probes do not isolate a single numerical cause.

Table 6: Diagnostics on all 12 completed contexts per target/modality arm. n counts contexts, and “Viol.” counts arms with at least one recorded pairwise-threshold violation between the two finest budgets. The remaining columns give maxima across those contexts: relative IG coordinate L_1 difference, finest-budget relative completeness residual, and maximum raw RMS response-curve change divided by the all-baseline RMS displacement, all in percent. RMS maxima include IG and mean-path-gradient rankings in both perturbation directions. Coordinate differences use the finer map’s L_1 norm. Comparators are 256 versus 512 for vision/language and 512 versus 1024 for state. These dependent checks neither estimate population error rates nor certify the accuracy of the finest maps. No setting or subset is approved.

Modality	Target	n	Viol.	IG L_1 (%)	Residual (%)	RMS (%)
Language	L	12	0	0.402	0.315	0.000
Language	Q	12	0	0.006	0.002	0.000
State	L	12	2	20.778	6.655	18.573
State	Q	12	2	32.119	11.436	1.641
Vision	L	12	4	1.234	0.288	7.361
Vision	Q	12	1	2.049	0.124	0.000

Repeatability and integration-budget sensitivity can occur together. A deterministic calculation can reproduce the same approximation at one budget and a different approximation at another. At the same time, coordinate errors can cancel when summed, leaving a small completeness residual. Neither repeatability nor one completeness statistic is enough to qualify a production setting. Appendix B shows why agreement across several budgets is also insufficient in general. Equal and opposite coordinate errors can preserve exact completeness while every selected coarse grid gives the same ranking and response curves, even though the ordering differs from exact IG. Correctness here means recovering the specified IG integral. Ranking quality under perturbations is a separate question, and this analytic possibility does not identify the cause of the observed RDT failures. The failures support a numerical diagnostic conclusion within the tested scope. Because the campaign is incomplete, they do not establish accuracy across modalities, tasks, checkpoints or future contexts.

7 Discussion and Limitations

We separate three questions: what the chosen scalar response measures, whether a numerical attribution is reliable, and whether its ranking contains useful information that depends on the model. Rescoring fixed recorded interventions addresses the first question because the response changes while the selected features stay fixed. It does not by itself establish numerical reliability or a useful model-dependent ranking. Our claims concern exact relationships and measurements reconstructed from the records. The numerical diagnostics apply only to their stated coverage. Establishing ranking efficacy would require a new paired study under common responses with informative controls. The present results do not establish it. The elementary identities show which operation changes a reported evaluation. They help avoid interpreting a response change as an empirical improvement in ranking, without proposing a new general theory of attribution. A common evaluation response defines the quantity being compared, but it is not a universal measure of explanation quality. Its action coordinates, units, baselines, perturbation groups and aggregation must be justified for the intended use. Matched random, input-difference and path-gradient controls would help separate information in the attribution from simple input structure and the IG multiplier. The protocols for these comparisons are deferred. They provide no comparative or learned-weight-specific result in this paper.

A practical evaluation procedure. We recommend four steps when reporting action-discrepancy attributions.

1. State the scalar attribution target and evaluation response separately. Also specify the reference action, fixed noise, feature representation, baseline, intervention groups and schedule.

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2. Compare rankings on authenticated matched contexts under each common response. Claims of efficacy need matched random and simple ranking controls. To assess a response change, rescore the same interventions, recompute the endpoint normalization and retain overshoots and undefined gaps.
 3. Assess repeatability and sensitivity of coordinates, rankings and responses across numerical settings on a declared context roster. Report failures and incomplete cases. Completeness and agreement on finitely many grids are numerical checks. They do not certify exact IG.
 4. Declare the unit of analysis and dependence grouping. Keep response sensitivity, numerical reliability and ranking quality distinct, because none alone establishes improved closed-loop behavior.

These steps make clear what each comparison identifies. They do not determine a universally adequate integration budget or evaluation response.

The historical records contain several sources of uncertainty, and each needs different treatment. Source-record hashes verify the released bytes, but exact checkpoint and context identities can remain unknown. Resampling whole episodes addresses dependence within episodes conditional on the recorded empirical mixture. It cannot repair shared or overwritten contexts, preprocessing confounds or missing training replicates. The released 1B results come from one checkpoint. The solver and alternative-target analyses cover fewer tasks than the verification campaign. Endpoint normalization permits unbounded response curves, so the results can be sensitive to endpoint gaps. Camera-token and language-token interventions differ in their group counts and meaning. Their AUCs therefore cannot be used interchangeably as measures of modality importance.

The overlay panels are retained as historical artifacts, not as evidence of successful grounding or as diagnoses of behavioral failure. The groups previously named from completeness residuals also describe numerical diagnostics. They are not independently validated labels for outcomes such as misgrounding or delayed credit. The original main-pass and early budget-extension numbers could not be recovered and are therefore absent from the revised primary tables. Appendix F reports only later budget cohorts that can be reconstructed. Those cohorts do not establish convergence.

8 Conclusion

An explicit auxiliary action-discrepancy score defines per-policy-call IG for a fixed-noise diffusion sampler. Across the saved RDT perturbation curves, changing only the evaluation response changes normalized AUC while keeping the selected features fixed. All eight reported paired cases have positive median changes, whereas overshooting tails make six means negative. Both the summary statistic and endpoint geometry therefore shape the conclusion. The finite-grid analytic counterexample rules out a universal random-ranking AUC of 0.5. The canonical retrospective tables keep measurements supported by the available records and exclude primary constants that could not be recovered. The actual-model diagnostics retain unresolved numerical failures and incomplete coverage. The smooth aliasing construction shows that finite grids can agree while producing an ordering different from exact IG. A common evaluation response is needed to interpret a comparison of ranking quality. These results do not show that one target or attribution method performs better under that response.

Ethical Statement

This research analyzes pretrained robotic manipulation policies in simulation. No human subjects were involved. Attribution is a diagnostic tool here, not a guarantee of safe robot deployment. Attractive maps, low completeness residuals or uncalibrated AUCs can create unjustified confidence if they are misinterpreted. The stated limits and controls are intended to prevent that inference.

Reproducibility Statement

The retrospective analysis parses a raw inventory authenticated by hashes, records decisions about repeated occurrences, and ties each result to physical source records. Revised tables and figures come from the same canonical artifact. Their points and intervals use identical populations, and a manuscript lineage registry identifies their sources. The artifact retains exclusions for conflicting keys, action-norm strata, endpoint and tail diagnostics, alternative episode weightings and conditional grouping sensitivities. The separate paired rescoring artifact includes its pre-computation analysis protocol, all eight cases, paired call and episode summaries, bootstrap membership, and an exact decomposition by saved-node geometry. A separately versioned exploratory supplement records every single-episode omission and its source identities without changing those primary results. The smooth nested-grid example separates its analytic proof from repeated FP32/FP64 CPU checks through the unchanged trapezoidal kernel. These operations reproduce arithmetic over saved records. They do not recover missing models, sidecars or original observations. The portable numerical projection contains the full planned roster, recorded checks and metrics, report hashes, and selected YCB FP32 tensors with their tensor hashes. It does not contain all raw artifacts for the 72 completed target-modality cells. Its CPU reproducer recomputes the reported diagnostics and saved-tensor checks from that projection. It does not repeat the complete original raw-artifact audit or execute the model again. Separately, the new runtime records immutable context, source, checkpoint, language, environment and noise identities. It fails explicitly on nonfinite attributions instead of sanitizing their coordinates. Neither a candidate PDF nor a successful script run alone establishes an authenticated scientific replication. Anonymous manuscript sources and a research supplement accompany this audit copy. The identified repository URL is omitted to support independent review.

References

- Julius Adebayo, Justin Gilmer, Michael Muelly, Ian J. Goodfellow, Moritz Hardt, and Been Kim. Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, pp. 9525–9536, 2018. URL <https://papers.nips.cc/paper/2018/hash/294a8ed24b1ad22ec2e7efea049b8737-Abstract.html>.
- Yotam Amitai, Yael Septon, and Ofra Amir. Explaining reinforcement learning agents through counterfactual action outcomes. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pp. 10003–10011, 2024. URL <https://ojs.aaai.org/index.php/AAAI/article/download/28863/29640>.
- Mariusz Bojarski, Anna Choromanska, Krzysztof Choromanski, Bernhard Firner, Larry Jackel, Urs Muller, P. Yeres, and Karol Zieba. VisualBackProp: Efficient visualization of CNNs for autonomous driving. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pp. 4701–4708, 2018. doi: 10.1109/ICRA.2018.8461053. URL <https://doi.org/10.1109/ICRA.2018.8461053>.
- Cheng Chi, Siyuan Feng, Yilun Du, Zhenjia Xu, Eric Cousineau, Benjamin Burchfiel, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. In *Proceedings of Robotics: Science and Systems (RSS)*, 2023. URL <https://roboticsproceedings.org/rss19/p026.html>.
- Samuel Greydanus, Anurag Koul, Jonathan Dodge, and Alan Fern. Visualizing and understanding Atari agents. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, pp. 1792–1801, 2018. URL <https://proceedings.mlr.press/v80/greydanus18a.html>.
- Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pp. 6840–6851, 2020. URL <https://papers.nips.cc/paper/2020/hash/4c5bcfec8584af0d967f1ab10179ca4b-Abstract.html>.
- Sara Hooker, Dumitru Erhan, Pieter-Jan Kindermans, and Been Kim. A benchmark for interpretability methods in deep neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, pp. 9734–9745, 2019. URL https://papers.nips.cc/paper_files/paper/2019/hash/fe4b855600d0f0cae99daa5c5c5a410-Abstract.html.

-
- Sumit Kumar Jha, Rickard Ewetz, Alvaro Velasquez, and Susmit Jha. On smoother attributions using neural stochastic differential equations. In *Proceedings of the 30th International Joint Conference on Artificial Intelligence (IJCAI)*, pp. 522–528, 2021. URL <https://www.ijcai.org/proceedings/2021/73>.
- Sumit Kumar Jha, Rickard Ewetz, Alvaro Velasquez, Arvind Ramanathan, and Susmit Jha. Shaping noise for robust attributions in neural stochastic differential equations. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pp. 9567–9574, 2022. URL <https://ojs.aaai.org/index.php/AAAI/article/view/21190>.
- Andrei Kapishnikov, Subhashini Venugopalan, Besim Avci, Ben Wedin, Michael Terry, and Tolga Bolukbasi. Guided integrated gradients: An adaptive path method for removing noise. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5050–5058, 2021. URL https://openaccess.thecvf.com/content/CVPR2021/html/Kapishnikov_Guided_Integrated_Gradients_An_Adaptive_Path_Method_for_Removing_Noise_CVPR_2021_paper.html.
- Jinwoong Kim and Sangjin Park. IMPACT-VLA: Interaction-aware multimodal propagation attribution via counterfactual trajectories for vision-language-action policies. *arXiv preprint arXiv:2609.15005*, 2026. URL <https://arxiv.org/abs/2609.15005>.
- Songming Liu, Lingxuan Wu, Bangguo Li, Hengkai Tan, Huayu Chen, Zhengyi Wang, Ke Xu, Hang Su, and Jun Zhu. RDT-1B: a diffusion foundation model for bimanual manipulation. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025. URL <https://arxiv.org/html/2410.07864v2>.
- Cheng Lu, Yuhao Zhou, Fan Bao, Jianfei Chen, Chongxuan Li, and Jun Zhu. DPM-Solver++: Fast solver for guided sampling of diffusion probabilistic models. *Machine Intelligence Research*, 22(4):730–751, 2025. URL <https://mi-research.net/article/doi/10.1007/s11633-025-1562-4>.
- Masahiro Makino, Yuya Asazuma, Shota Sasaki, and Jun Suzuki. The impact of integration step on integrated gradients. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: Student Research Workshop*, pp. 279–289. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.eacl-srw.22. URL <https://aclanthology.org/2024.eacl-srw.22/>.
- Octo Model Team, Dibya Ghosh, Homer Walke, Karl Pertsch, Kevin Black, Oier Mees, et al. Octo: An open-source generalist robot policy. In *Proceedings of Robotics: Science and Systems (RSS)*, 2024. URL <https://www.roboticsproceedings.org/rss20/p090.pdf>.
- Vitali Petsiuk, Abir Das, and Kate Saenko. RISE: Randomized input sampling for explanation of black-box models. In *Proceedings of the British Machine Vision Conference (BMVC)*, 2018. URL <https://www.bmvc2018.org/contents/papers/1064.pdf>.
- Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020. URL <https://jmlr.org/papers/v21/20-074.html>.
- Yao Rong, Tobias Leemann, Vadim Borisov, Gjergji Kasneci, and Enkelejda Kasneci. A consistent and efficient evaluation strategy for attribution methods. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, pp. 18770–18795, 2022. URL <https://proceedings.mlr.press/v162/rong22a.html>.
- Haoyuan Shi, Xiancong Ren, Yingji Zhang, Qinfan Zhang, Jiayu Hu, Haozhe Shan, Han Dong, Jinpeng Lu, Yinda Chen, Yi Zhang, Yong Dai, and Xiaozhu Ju. VLA-Trace: Diagnosing vision-language-action models through representation and behavior tracing. *arXiv preprint arXiv:2605.30117*, 2026. URL <https://arxiv.org/abs/2605.30117>.
- Shree Singhi and Anupriya Kumari. Strengthening interpretability: An investigative study of integrated gradient methods. *arXiv preprint arXiv:2409.09043*, 2024. doi: 10.48550/arXiv.2409.09043. URL <https://arxiv.org/abs/2409.09043>.

-
- Mukund Sundararajan, Ankur Taly, and Qiqi Yan. Axiomatic attribution for deep networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, pp. 3319–3328, 2017. URL <https://proceedings.mlr.press/v70/sundararajan17a.html>.
- Raphael Tang, Linqing Liu, Akshat Pandey, Zhiying Jiang, Gefei Yang, Karun Kumar, Pontus Stenetorp, Jimmy Lin, and Ferhan Ture. What the DAAM: Interpreting Stable Diffusion using cross attention. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 5644–5659, 2023. URL <https://aclanthology.org/2023.acl-long.310/>.
- Stone Tao, Fanbo Xiang, Arth Shukla, Yuzhe Qin, Xander Hinrichsen, Xiaodi Yuan, Chen Bao, Xinsong Lin, Yulin Liu, Tse-kai Chan, et al. ManiSkill3: GPU parallelized robotics simulation and rendering for generalizable embodied AI. *arXiv preprint arXiv:2410.00425*, 2024. URL <https://arxiv.org/abs/2410.00425>.
- Richard Tomsett, Dan Harborne, Supriyo Chakraborty, Prudhvi Gurram, and Alun D. Preece. Sanity checks for saliency metrics. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pp. 6021–6029, 2020. URL <https://ojs.aaai.org/index.php/AAAI/article/view/6064>.
- Chase Walker, Sumit Kumar Jha, Kenny Chen, and Rickard Ewetz. Integrated decision gradients: Compute your attributions where the model makes its decision. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pp. 5289–5297, 2024. URL <https://ojs.aaai.org/index.php/AAAI/article/view/28336>.
- Chase Walker, Md Rubel Ahmed, Sumit Kumar Jha, and Rickard Ewetz. Explaining ViTs using information flow. In *Proceedings of the International Conference on Artificial Intelligence and Statistics (AISTATS)*, pp. 2440–2448, 2025a. URL <https://proceedings.mlr.press/v258/walker25a.html>.
- Chase Walker, Sumit Kumar Jha, and Rickard Ewetz. Metric-driven attributions for vision transformers. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025b. URL <https://openreview.net/pdf?id=rGP2jbWt0L>.
- Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. Sigmoid loss for language image pre-training. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 11975–11986, 2023. URL https://openaccess.thecvf.com/content/ICCV2023/html/Zhai_Sigmoid_Loss_for_Language_Image_Pre-Training_ICCV_2023_paper.html.
- Hanxin Zhang, Mingshuo Xu, Abdulqader Dhafer, Shigang Yue, Hongbiao Dong, and Zhou Daniel Hao. Embodied interpretability: Linking causal understanding to generalization in vision-language-action models. *arXiv preprint arXiv:2605.00321*, 2026. URL <https://arxiv.org/abs/2605.00321>.
- Brianna Zitkovich, Tianhe Yu, Sichun Xu, Peng Xu, Ted Xiao, Fei Xia, et al. RT-2: Vision-language-action models transfer web knowledge to robotic control. In *Proceedings of the 7th Conference on Robot Learning (CoRL)*, pp. 2165–2183, 2023. URL <https://proceedings.mlr.press/v229/zitkovich23a.html>.

A Response-Geometry Derivations

The identities in Section 3.4 keep the action reference fixed and assume that the relevant path gradients are integrable. Differentiating $Q = -r/(2D\sigma^2)$ and $L = -\sqrt{r + \epsilon}$ gives Equation 5. To obtain Equation 11, set $a_0 = \sqrt{\epsilon}$, $b = \sqrt{R + \epsilon}$ and $z = \sqrt{r + \epsilon}$. Then $R = b^2 - a_0^2$ and $r = z^2 - a_0^2$, so rationalization gives

$$\frac{b^2 - z^2}{R} - \frac{b - z}{b - a_0} = \frac{(z - a_0)(b - z)}{R}. \quad (12)$$

For in-range residuals, the maximum difference is $(b - a_0)^2/(4R) \leq 1/4$. This pointwise derivation applies to interventions from a fixed ranking. It does not order different rankings or establish improvements in task success.

For the rank-reversal example, take $c = 4/5$ and parameterize the path by (t, t) . The scalar residual is $d(t) = 1 + c - t^2 - ct = (1 - t)(1 + t + c) \geq 0$. The quadratic path integrands are $2td(t)$ and $cd(t)$, giving

$$\text{IG}^Q = \left(\frac{1}{2} + \frac{c}{3}, \frac{2c}{3} + \frac{c^2}{2} \right) = \left(\frac{23}{30}, \frac{64}{75} \right). \quad (13)$$

For the stabilized negative norm, the path integrands are $2td(t)/\sqrt{d(t)^2 + \epsilon}$ and $cd(t)/\sqrt{d(t)^2 + \epsilon}$. Using the bound $d(t) \geq c(1 - t)$ and monotonicity of $1 - d/\sqrt{d^2 + \epsilon}$ gives

$$\int_0^1 \left(1 - \frac{d(t)}{\sqrt{d(t)^2 + \epsilon}} \right) dt \leq 1 - \frac{\sqrt{c^2 + \epsilon} - \sqrt{\epsilon}}{c} \leq \frac{\sqrt{\epsilon}}{c}. \quad (14)$$

At the implemented stabilizer, this gives $0 \leq 1 - \text{IG}_1^L \leq 2\sqrt{\epsilon}/c = 2.5 \times 10^{-6}$ and $0 \leq c - \text{IG}_2^L \leq \sqrt{\epsilon} = 10^{-6}$. All coordinates are positive. The stabilized L margin is at least 0.1999975, while Q has the reverse ordering with exact margin 13/150. The artifact checks these identities through independent differentiation, integration and normalization. The checks include overshoots, zero gaps and a policy example without rank reversal.

B A Smooth Counterexample to Nested-Grid Agreement

Finitely many integration grids can agree while giving inaccurate coordinate integrals. Exact completeness and unchanged perturbation curves do not rule this out. This construction shows why without a model experiment. All equalities in the construction are statements in real arithmetic.

For a positive integer M , define the smooth scalar policy and target

$$h(t) = \frac{2}{5} \sin^2(\pi Mt), \quad (15)$$

$$\mu(x, y) = \frac{2}{5}x + \frac{3}{5}y + (x - y)h\left(\frac{x + y}{2}\right), \quad (16)$$

$$Q(x, y) = -\frac{1}{2}(\mu(x, y) - 1)^2. \quad (17)$$

Take baseline $(0, 0)$, actual input $(1, 1)$ and the fixed own-reference action $\mu(1, 1) = 1$. This has the same quadratic self-reference form as the main target, with one active action entry and $\sigma^2 = 1$. Along the straight path (t, t) , the transverse term in μ vanishes. Thus $\mu(t, t) = t$, $Q(t, t) = -(1 - t)^2/2$, and

$$\nabla Q(t, t) = (1 - t) \left(\frac{2}{5} + h(t), \frac{3}{5} - h(t) \right). \quad (18)$$

The endpoint score gap is $1/2$. Both coordinate integrals are positive, but their exact values differ.

Let T_m denote componentwise composite trapezoidal integration with m intervals. If m divides M , every node $t = k/m$ has $h(t) = 0$. The remaining coordinate functions are linear, giving

$$T_m = \left(\frac{1}{5}, \frac{3}{10} \right), \quad \mathbf{1}^\top T_m = \frac{1}{2} = Q(1, 1) - Q(0, 0). \quad (19)$$

Every such grid has zero completeness residual and the same strict absolute-coordinate ranking, with feature 2 first. Under a fixed deterministic ranking rule, this order gives identical insertion and deletion masks. Evaluating them on the same policy therefore produces identical response curves, even though the coordinate integrals are inaccurate.

At the same time, $h(1-t) = h(t)$ and $\int_0^1 h(t) dt = 1/5$, so

$$\int_0^1 (1-t)h(t) dt = \frac{1}{10}, \quad \text{IG}(Q) = \left(\frac{3}{10}, \frac{1}{5} \right). \quad (20)$$

The exact ranking places feature 1 first. The coarse-grid errors are $(-1/10, 1/10)$. They cancel in the sum, but their coordinate L_1 norm is $1/5$, or 40% of the exact map’s L_1 norm. The path score is a quadratic polynomial. The grids miss how its gradient is divided between coordinates, which a scalar completeness check cannot recover.

For this construction, the $2M$ grid resolves the discrepancy. Its sampled h values alternate between 0 and $2/5$, giving $T_{2M}h = 1/5$. Grid symmetry gives $T_{2M}[(1-t)h] = \frac{1}{2}T_{2M}h = 1/10$, recovering Equation 20. This does not make doubling a universally sufficient rule. For any fixed finite list of uniform interval counts, choosing M as a common multiple places every grid in Equation 19. The example provides no derivative bound to control the variation that these grids miss.

Exact integration and perturbation quality are different questions. For the displayed $M = 128$, $h(1/2) = 0$, so whole-feature interventions give $\mu(1, 0) = 2/5$ and $\mu(0, 1) = 3/5$. With the same normalized Q response and grid $\{0, 1/2, 1\}$, the aliased feature-2-first ordering has insertion ordinates $(0, 0.84, 1)$ and deletion ordinates $(1, 0.64, 0)$. These give AUCs 0.67 and 0.57. The exact-IG feature-1-first ordering instead has insertion AUC 0.57 and deletion AUC 0.67. The inaccurate map’s ranking therefore scores better on both conventional criteria in this toy: larger insertion AUC and smaller deletion AUC. This happens because the finite off-path interventions measure a different property from the diagonal path integral. The transverse policy term changes the coordinate gradients without changing the Boolean intervention values. Neither completeness nor exact integration guarantees an optimal perturbation score. This separation is specific to the toy and does not establish RDT ranking efficacy.

The standalone artifact `analysis/revision/nested_grid_aliasing.py` evaluates this CPU toy through the unchanged production trapezoidal IG kernel with $M = 128$, in FP32 and FP64. Budgets 32, 64, 128 give the incorrect order, while 256 reproduces the analytic order to numerical tolerance. The report retains measured coordinate errors, completeness residuals, repeated-run equality and common response curves. It does not replace finite-precision results with the exact equalities above. The two-coordinate example uses no RDT weights or observations. It shows why finite-grid agreement alone cannot certify the integral. It does not explain why any particular PickSingleYCB diagnostic failed or approve a production integration setting.

C Secondary Historical Results

These analyses preserve additional observations from the saved campaigns. Limits on training, reference, context and preprocessing prevent conclusions about model-scale effects, learned-weight-specific efficacy or a causal effect of denoiser contraction.

C.1 The Separately Recorded 1B Cohort

The saved 1B campaign shows why episode weighting matters. Changing the weights changes the quantity being estimated. Table 7 uses the same 1,199 calls from 60 episodes for both summaries. Weighting affects the language summaries more than the vision summaries in this cohort. The cohort also mixes language-baseline provenance. Neither estimate comes from replicated training, and neither establishes a controlled improvement over 170M.

Table 7: Results from the saved RDT-1B PickCube campaign. Both quantities use 1,199 source calls and 60 episodes. The first gives equal weight to calls, while the second gives equal weight to episode medians. The conditional 95% episode intervals describe this released mixture. The evaluation seeds share a checkpoint, and the pooled language population includes two recorded baseline provenances.

Estimand	Modality	Insertion AUC	Deletion AUC
Median over calls	Vision	0.940 [0.926, 0.954]	0.288 [0.250, 0.312]
Median over calls	Language	0.541 [0.449, 0.605]	0.267 [0.187, 0.327]
Median of episode medians	Vision	0.940 [0.927, 0.959]	0.289 [0.235, 0.313]
Median of episode medians	Language	0.488 [0.368, 0.581]	0.297 [0.151, 0.342]

C.2 Randomization and Baseline Diagnostics

Table 8 reconstructs the standard verification-campaign correlations. Substantial rank association persists after model randomization, while input shuffling changes the maps more strongly. This contrast does not show that model dependence is unnecessary or establish semantic grounding. A map based entirely on input magnitude can change under input shuffling while ignoring learned policy weights. Appendix E reports the saved frozen-reference and cascade variants, along with a sensitivity analysis of dependence grouping.

Table 8: Month 7 model randomization (C1) and input shuffle (C2) results use all recorded calls. Values are median Spearman correlations with conditional 95% episode intervals. The checks measure different forms of dependence. Low C2 correlation does not establish dependence on learned model weights, while C1 changes the integration budget as well as the weights.

Task	Check	Calls	Episodes	Vision ρ	Language ρ
PegInsertion	C1	100	16	0.960 [0.949, 0.969]	0.959 [0.947, 0.966]
PegInsertion	C2	100	16	0.089 [0.067, 0.111]	0.018 [0.004, 0.036]
PickCube	C1	100	26	0.956 [0.948, 0.961]	0.926 [0.917, 0.944]
PickCube	C2	100	26	0.142 [0.128, 0.157]	0.117 [0.099, 0.150]
PickYCB	C1	100	26	0.943 [0.930, 0.949]	0.917 [0.903, 0.931]
PickYCB	C2	100	26	0.164 [0.149, 0.175]	0.128 [0.081, 0.155]
StackCube	C1	100	26	0.958 [0.952, 0.965]	0.965 [0.960, 0.969]
StackCube	C2	100	26	0.140 [0.117, 0.154]	0.266 [0.246, 0.279]

The IG multiplier ($x_i - x'_i$) can create shared structure, but it does not determine a particular ranking. Gradients along the path can dominate. The input features also come from retained learned encoders/adaptors. Persistent correlations therefore motivate input-only and path-gradient-only controls, but do not establish that the IG multiplier dominates. Spearman correlation cannot be read as the fraction of model-independent information. The baseline comparisons in Appendix E show imperfect agreement. Too few groups are available to establish equivalence.

C.3 Action Displacement and Solver Resolution

The quadratic score is a normalized squared displacement, so its numerical magnitude depends directly on the chosen convention. Dropping $1/D$ scales exact scores and attributions by D , while exact ranks, normalized AUC and relative completeness remain unchanged. Whether an action change matters operationally requires action units and a justified behavioral criterion. The algebra alone cannot answer that question.

The saved solver analysis compares median relative action displacement under top-5% vision deletion between $T = 2$ and $T = 20$. Figure 2 reports this endpoint contrast by conditionally resampling nominal episode pairs. Reference actions vary across solver settings, and exact contexts/noise cannot be authenticated. The intervals allow appreciable increases as well as decreases. Changing T changes the discretization of the full

denoising horizon. It does not add steps along one fixed trajectory. These observations therefore neither establish a causal contraction mechanism nor rule one out.

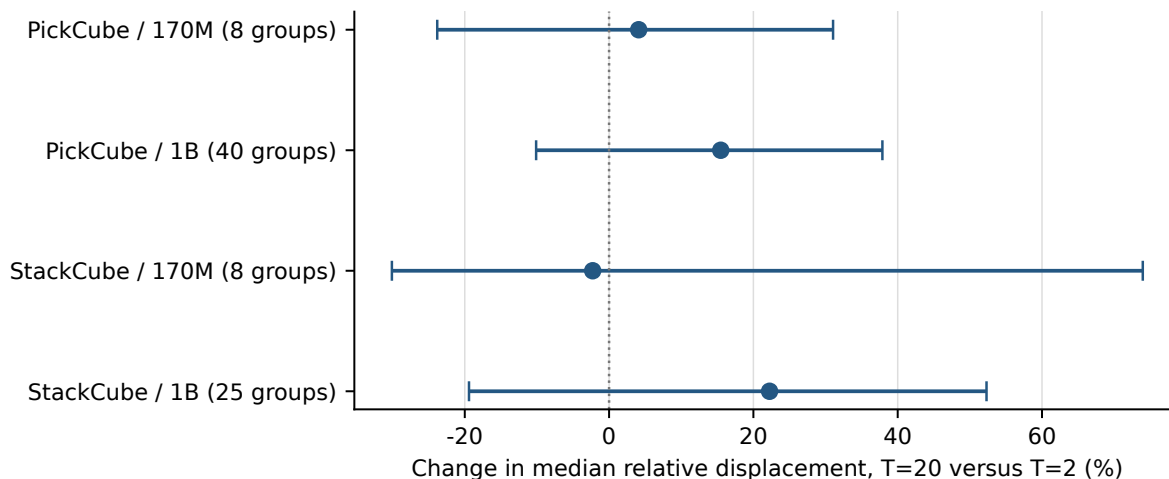


Figure 2: Retrospective change in median relative displacement under top-5% vision deletion. The intervals conditionally resample nominal episode pairs across solver resolutions and apply to the released records and grouping. The vertical reference marks no endpoint change. It is neither a contraction test nor an equivalence margin. Raw context/noise identity remains unknown.

D Exploratory Episode Influence

Large finite response changes affect the paired summaries in Table 2. To identify this sensitivity, we omit each of the 30 recorded episodes in turn, use all calls from the other 29 episodes, and recompute their equally weighted episode mean. Table 9 gives every case. The analysis followed an exploratory audit of the same records. Its separately saved protocol is therefore not an outcome-blind preregistration. The omission ranges describe deterministic influence. They are not confidence intervals and do not establish that the recorded episodes form an independent sample. The full untrimmed means, medians and conditional intervals remain unchanged.

Table 9: Exploratory omissions of one complete recorded episode. The range covers all 30 omission means. “Same sign” counts agreement with the full-cohort mean. Each case starts with 750 calls in 30 episodes, and rows within a ranking cohort reuse calls. These ranges are not confidence intervals. The separate influence artifact supplies all 240 omission records and their retained-population hashes.

Ranking	Modality	Curve	Full mean	Omission-mean range	Same sign
Q	vision	insertion	-17.0966	[-17.6910, -11.5667]	30/30
Q	vision	deletion	0.1227	[0.1208, 0.1272]	30/30
Q	lang	insertion	-0.1819	[-0.1937, -0.0932]	30/30
Q	lang	deletion	-0.0095	[-0.0158, 0.0182]	27/30
L2	vision	insertion	-18.5620	[-19.2071, -11.9674]	30/30
L2	vision	deletion	0.0184	[0.0128, 0.1264]	30/30
L2	lang	insertion	-0.0329	[-0.0399, -0.0193]	30/30
L2	lang	deletion	-1.3710	[-1.4246, 0.0328]	29/30

All four negative insertion means stay negative under every individual omission. The Q language-deletion mean changes sign in 3 of 30 omissions, and the L2 language-deletion mean changes sign in 1 of 30. Both already have conditional bootstrap intervals spanning zero. In the latter case, the most negative episode

accounts for 97.29% of the sum of magnitudes of negative episode means. This describes concentration among negative episode means, not a share of the signed net mean. It does not justify excluding that episode.

Endpoint normalization helps explain the extreme vision-insertion changes. For the most negative Q-ranking call, the baseline squared residual is 0.0174195, the smallest value among its 750-call cohort. The same call’s maximum saved perturbed residual is 653.053, a ratio of about 37,490. The finite Q endpoint gap is 1.7011×10^{-5} , above the historical 10^{-9} cutoff. Its native Q AUC is -4464.9082 , while its L2 rescore is -26.0696 . The call therefore combines a small baseline discrepancy with a large intervention response. The division is neither undefined nor by a machine-zero value. These quantities do not calibrate physical importance in action units. The artifact supplies the same diagnostics within each separate ranking cohort. Matching nominal episode labels cannot authenticate a cross-arm pair.

E Additional Retrospective Diagnostics

Both the frozen-reference and backbone-cascade analyses retain learned encoders/adaptors. Their historical context identities remain uncertain. Table 10 reports recorded-episode grouping and conditional shared-context grouping. Comparing them measures sensitivity to grouping, but does not establish that either grouping is independent. Table 11 gives point summaries of the baseline rankings. The small number of groups prevents an equivalence conclusion.

Table 10: Results for the saved backbone-randomization variants. The shared-context grouping joins nominal task and episode keys across recorded seeds to check sensitivity to dependence. It does not establish that the inputs were shared. Intervals are conditional on each stated grouping. Learned encoders and adaptors remain in place, while sidecar reuse and language-baseline deviations remain unresolved.

Variant	Grouping	Calls	Groups	Vision ρ	Language ρ
Frozen	Recorded episodes	400	16	0.929 [0.905, 0.940]	0.862 [0.835, 0.879]
Frozen	Shared-context sensitivity	400	8	0.929 [0.904, 0.947]	0.862 [0.820, 0.886]
Cascade	Recorded episodes	400	16	0.713 [0.696, 0.738]	0.910 [0.896, 0.920]
Cascade	Shared-context sensitivity	400	8	0.713 [0.693, 0.763]	0.910 [0.894, 0.926]

Table 11: We compare correlations between saved baseline rankings. For each task there are only two nominal episode groups. The artifact reports intervals to describe the resampling results, but we do not present them as calibrated uncertainty. A positive correlation alone does not show that the baselines can be used interchangeably.

Task	Calls	Groups	Black/gray	Black/blur	Gray/blur
PickCube	30	2	0.825	0.539	0.455
StackCube	30	2	0.814	0.546	0.496

F Saved Integration-Budget Cohorts

Table 12 replaces the earlier table of unrecovered and duplicated constants with later cohorts that can be reconstructed. This historical sensitivity analysis explicitly retains the norm-selected population and occurrence rule. These calls are not paired with the earlier $m = 64$ main pass on identical observations, base-lines, runtime and noise. Truncating trajectories to match call depth does not create such pairs. Assessing an integration-budget effect therefore requires authenticated fixed contexts, with precision and quadrature controlled.

Table 12: Results for the $m = 128$ cohorts that can be reconstructed. The intervals are conditional on the recorded cohorts and do not show convergence under paired contexts. The final row omits every conflicting repeated key. The missing original PickCube and StackCube subsets have not been reconstructed or replaced with constants.

Task	Seed	Calls	Groups	Vision residual $\leq 3\%$ (%)
PegInsertion	42	189	8	64.0 [50.3, 76.9]
PegInsertion	142	160	7	76.2 [67.1, 85.0]
PickYCB	42	194	8	73.7 [62.1, 85.5]
PickYCB	142	195	8	71.8 [60.9, 83.1]
PegInsertion (conflicts excluded)	142	142	7	75.4 [63.8, 85.0]